

Robust Attitude Controller Design for Bi-copter on SO(3)

Tianyou Mao, Bosong Duan, and Chuangqiang Guo* 

Abstract: The bi-copter features a compact structural layout and energy-efficient drive configuration, enabling it to perform tasks that some quadrotors cannot handle. However, research on control methods for bi-copters remains underdeveloped. The system's inherent non-minimum phase characteristics, underactuated nature, and nonlinearity make attitude control particularly challenging. In this study, a robust attitude controller based on the three-dimensional special orthogonal group (SO(3)) is designed for the bi-copter. The proposed controller enables the aircraft to track attitude signals with inter-axis coupling and rapid variations, while also offering disturbance rejection capabilities. The stability of the controller has been demonstrated, and its effectiveness validated through both simulations and prototype testing. This more comprehensive control strategy expands the bi-copter's feasible attitude range beyond the vicinity of hover, thereby broadening its potential applications.

Keywords: Attitude anti-disturbance control, bi-copter, controller design in SO(3), underactuated systems.

1. INTRODUCTION

In recent years, multi-rotor drones have experienced rapid development due to their broad range of applications, and are now widely used in areas such as search and rescue [1,2], data collection and monitoring [3,4], agriculture [5,6], and formation control [7-9]. Typical multi-rotor drones generally feature four or more rotors, with attitude control achieved by varying the rotational speed of the fixed-position propellers. While this mechanical structure is simple and reliable, it also presents challenges, such as high power consumption and difficulties in optimizing lateral dimensions [10,11]. In contrast, bi-copter drones offer significant advantages by reducing power requirements and mitigating typical adverse effects during flight [12], making them one of the most energy-efficient designs under size constraints [13]. Theoretically, bi-copters consume 30% [10] less power than quadcopters of the same takeoff weight and size, allowing for greater payload capacity and longer endurance during missions. Additionally, the bi-copter configuration results in a significant reduction in horizontal dimensions, enhancing its maneuverability in confined spaces and enabling it to navigate through narrower areas [13]. These advantages make the bi-copter potentially suitable for tasks that quadcopters may not easily perform.

But at the same time, the bi-copter configuration presents significant challenges for controller design. It is a typical underactuated, nonlinear system [14], with cou-

pled control variables. Due to the torque-induced effects from the servo drive mechanisms, the bi-copter exhibits pronounced non-minimum phase characteristics [13,15], which degrade the pitch axis response [16]. Moreover, while this configuration reduces the drone's horizontal dimensions, it inevitably increases the vertical size when a certain payload capacity is required. This directly results in a higher moment of inertia for pitch and roll motions, further reducing the controller's responsiveness. Consequently, the bi-copter remains an immature technology and is not yet a stable flight platform for practical applications. Most current research still focuses on basic aspects such as hover control and navigating narrow gaps [16], with limited development in specialized flight capabilities. For instance, there is a lack of studies on the bi-copter's response to attitude signals with inter-axis coupling and high-speed variations. Such capabilities could significantly enhance performance in multi-modal aircraft [17-19] developed from dual-propeller configurations, enabling smoother transitions between different flight modes, though they also pose considerable challenges for control strategies.

It is well known that the control methods based on linearized approximations of Euler angles only achieve good control performance when the drones' attitude is close to the hover point. When the desired attitude angles change rapidly or multiple axes change simultaneously, feedback control needs to be applied directly to the quaternion or attitude matrix. The primary drawback of unit quaternions

Manuscript received March 18, 2025; revised July 22, 2025; accepted September 12, 2025. Recommended by Associate Editor Hyeonbeom Lee under the direction of Senior Editor Dongjun Lee.

Tianyou Mao and Chuangqiang Guo are with the State Key Laboratory of Robotics and System, Harbin Institute of Technology, Harbin 150001, Heilongjiang Province, China (e-mails: 23b908026@stu.hit.edu.cn, chuangqiang.guo@hit.edu.cn). Bosong Duan is with the School of Mechanical Engineering, Zhejiang University, Hangzhou 310027, Zhejiang Province, China (e-mail: bosongduan@zju.edu.cn).

* Corresponding author.

is the potential occurrence of the unwinding phenomenon [20], which may require additional corrective measures [21]. In contrast, control methods directly based on SO(3) do not encounter such issues. In recent years, several studies [22-24] have applied these methods to quadrotor control. However, these state variables are less intuitive than Euler angles, making controller design more challenging. Even the seemingly simple PID controllers require complex derivations in SO(3), as shown in the research by Yu *et al.* [25]. Moreover, the performance of traditional PID controllers [14,26,27] or cascade PID controllers [10,28] is highly dependent on the precision of manual parameter tuning. The nonlinear coupling between the three rotational axes makes the tuning process very difficult. Although optimization methods such as genetic algorithms (GA) [29,30] or particle swarm optimization (PSO) [13] can yield optimal parameters, they impose stringent demands on the precision of dynamic modeling and exhibit high computational resource requirements. Parameters derived from simulation models are often not directly applicable to actual experimental environments. Therefore, such methods typically serve only as a supplementary approach for parameter adjustment. More importantly, when the model contains uncertainties, the stability margin of these controllers is entirely unknown [15], and their disturbance rejection capabilities may not be guaranteed. The H_∞ robust controller performs well in dealing with severe external disturbances or center-of-mass shifts in bi-copter systems [15,31]. However, it requires the use of system identification methods to establish an accurate control model in advance, which is a highly complex task. Additionally, this type of controller suffers from the issue of high order [15], making it difficult to apply in controllers based on SO(3). The sliding mode controller (SMC) [32] also faces challenges in designing sliding surfaces within Lie groups, which may become overly complex and theoretically difficult to analyze. In contrast, the backstepping (BKS) controller [33] offers two distinct advantages: easier Lyapunov stability proof and better compatibility with the geometric control framework of SO(3) manifolds. But a standalone BKS controller struggles to handle uncertainties in the system model and is only suitable for ideal flight conditions. The majority of existing finite-time control methods encounter similar issues and impose strict constraints on the system's initial state [34,35]. To address parameter uncertainties and potential external disturbances in bi-copter attitude control, the extended state observer (ESO) may offer an effective and convenient solution [36]. Numerous studies [37-40] have demonstrated the value of ESO in multi-rotor drones controller design. This observer-based controller does not require an accurate system model and offers more convenient tuning methods [41], making it easier to implement.

Against this background, this study designs a highly robust attitude controller for bi-copters operating on SO(3).

The primary contributions of this research include:

- 1) The expression of the ESO in SO(3) is derived, and a BKS controller based on the Lyapunov method is also designed within the same framework. The stability of the observer, controller, and the entire control system has been proven.
- 2) Simulations and experimental tests show that the proposed control method enables the bi-copters to track attitude commands with cross-axis coupling and large-range high-speed variations while maintaining certain disturbance rejection capability.
- 3) This more comprehensive control approach expands the controllable attitude range of the bi-copters beyond the vicinity of the hover point, broadening its potential application prospects.

The rest of the paper is organized as follows: Section 2 introduces a bi-copter dynamic modeling method that considers the non-minimum phase characteristics. Section 3 presents the overall control architecture, and designs the ESO and BKS controllers for the bi-copter system within SO(3). The bounded stability of the controllers is proven using the Lyapunov method. Section 4 compares the response characteristics of several controllers, validates the robustness of the controller designed in this study, and discusses how the proposed control theory can be applied to a practical bi-copter prototype. Finally, Section 5 provides the conclusion.

2. DYNAMIC MODELS

Firstly, the parameters and coordinate systems of the bi-copter are defined in Fig. 1. $O_gX_gY_gZ_g$ represents the ground coordinate system, using the East-North-up (ENU) definition. $O_bX_bY_bZ_b$ denotes the body coordinate system, where O_b coincides with the center of mass of the bi-copter. The Z_b -axis coincides with the body's longitudinal axis, with the positive direction pointing towards the nose of the aircraft. The X_b -axis is perpendicular to the line connecting the two propellers and points towards the front of the body. The Y_b -axis is parallel to the line connecting the two propellers and points towards the left side of the body. F_1 and F_2 represent the thrust forces generated by the left and right propellers, respectively, while ξ_1 and ξ_2 denote the angles between the thrust vector directions and the Z_b -axis. ϕ represents the roll angle, θ represents the pitch angle, and ψ represents the yaw angle.

Since this study aims to propose a control method for a specific type of spatial rotational motion, the treatment of translational motion is not addressed in the dynamics modeling and controller design. This is because, at this stage, the bi-copter is not suitable for position planning under conditions of multi-axis large-angle deflections. Mature solutions for position planning near the hover point are already available for [14,33].

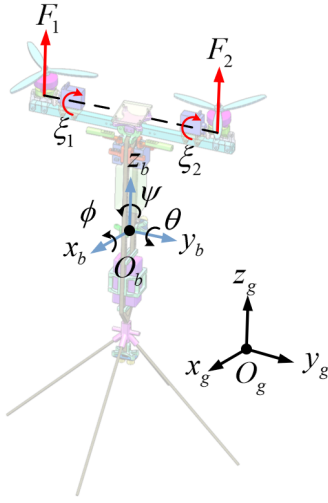


Fig. 1. Rigid body simplified dynamic model of the bi-copter.

To clarify the relationship between the control output and the torque of the bi-copter, the control quantity can be defined as

$$\begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix} = \begin{bmatrix} F_1 \sin \xi_1 + F_2 \sin \xi_2 \\ F_1 \sin \xi_1 - F_2 \sin \xi_2 \\ F_1 \cos \xi_1 + F_2 \cos \xi_2 \\ F_1 \cos \xi_1 - F_2 \cos \xi_2 \end{bmatrix}. \quad (1)$$

The transformation relationship between the control quantity and the actual control output is given by

$$\begin{bmatrix} \xi_1 \\ \xi_2 \\ F_1 \\ F_2 \end{bmatrix} = \begin{bmatrix} \arctan \left(\frac{U_1 + U_2}{U_3 + U_4} \right) \\ \arctan \left(\frac{U_1 - U_2}{U_3 - U_4} \right) \\ \frac{U_3 + U_4}{2 \cos \xi_1} \\ \frac{U_3 - U_4}{2 \cos \xi_2} \end{bmatrix}. \quad (2)$$

The torque generated by the control output in the body coordinate system is expressed as

$$\tau_c = L_i [U_4, -U_1, U_2], \quad (3)$$

where L_i is the diagonal matrix of the arm length. Based on the Euler equations, the dynamic equations of spatial rotation can be derived as

$$J_b \dot{\omega}_b + \omega_b \times J_b \omega_b = \tau_c + \tau_r + \tau_d, \quad (4)$$

where J_b is the diagonal matrix of the body's moment of inertia, ω_b is the angular velocity in the body coordinate system, and τ_d is the torque induced by disturbances. τ_r is the reaction torque caused by the inertia of the propellers and motors, which is the main cause of non-minimum

phase characteristics. It can be expressed as

$$\tau_r = L_i \sum_{i=1}^2 \kappa_i R_{pb}(\xi_i) m_i^p r_i^p \ddot{\xi}_i, \quad (5)$$

where κ_i is the direction vector, $R_{pb}(\xi_i)$ is the 2D rotation matrix from the propeller and motor to the body coordinate system, m_i^p is the mass of a set of propellers and motors, and r_i^p is the radius of rotation of the propellers and motors. These can be expressed as

$$\kappa_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \\ -1 & 0 \end{bmatrix}, \quad \kappa_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 0 \end{bmatrix}, \quad (6)$$

$$R_{pb}(\xi_i) = \begin{bmatrix} \cos \xi_i & -\sin \xi_i \\ \sin \xi_i & \cos \xi_i \end{bmatrix}. \quad (7)$$

Additionally, the rotational dynamics equation for the body on SO(3) can be expressed as [25]

$$\dot{R} = R \cdot \text{sk}(\omega_b), \quad (8)$$

where R is the rotation matrix that transforms from the ground coordinate system to the body coordinate system, and the mapping relationship $\text{sk}(\cdot)$ is defined as

$$\text{sk}(\Phi) = \begin{bmatrix} 0 & -\Phi_3 & \Phi_2 \\ \Phi_3 & 0 & -\Phi_1 \\ -\Phi_2 & \Phi_1 & 0 \end{bmatrix}, \quad (9)$$

where $\Phi = [\Phi_1, \Phi_2, \Phi_3]$ represents an arbitrary three-dimensional vector. The inverse mapping of $\text{sk}(\cdot)$ is represented as $(\cdot)^\vee$.

3. CONTROLLER DESIGN AND STABILITY ANALYSIS

This section introduces the design method of the BKS controller and ESO in SO(3). The overall block diagram of the attitude control theory is shown in Fig. 2. For the sake of intuition, the desired attitude is typically defined in Euler angles. In the control system, the desired Euler angles and angular velocities are first converted into the rotation matrix R_d in SO(3) and its derivative $\dot{R}_d$. Then, the error matrices R_e and $\dot{R}_e$ between the desired and actual attitude matrices are computed. In the BKS controller, the desired control torque τ_c is designed using the Lyapunov method. The ESO is primarily designed to compensate for system uncertainties and external disturbances. In the simulation environment, the actual rotation matrix R can be computed based on the bi-copter's dynamics equation. In practical prototype systems, if certain body angular velocities are difficult to measure directly, ESO can be used to estimate them. In the case presented in this paper, since the body angular velocity can be directly obtained from the IMU, the role of the ESO is primarily to observe disturbances.

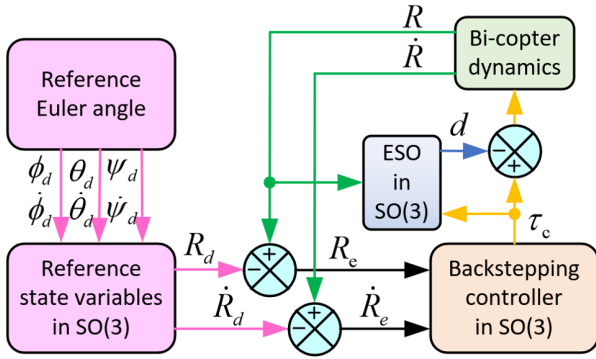


Fig. 2. Overall block diagram of the attitude control.

3.1. Design and stability analysis of ESO

In practical engineering control problems, differential calculations are highly sensitive to noise, making it inconvenient to balance the effects of reaction torque by adding a feedforward loop. Therefore, during the design of ESO, τ_r can be treated as an internal disturbance without requiring additional compensation. Equation (4) is primarily used for constructing the simulation model rather than for the controller design. Thus, the system's dynamic model can be expressed as

$$\begin{cases} J_b \dot{\omega}_b + \omega_b \times J_b \omega_b = \tau_c + d, \\ \dot{R} = R \cdot \text{sk}(\omega_b), \end{cases} \quad (10)$$

where d is the additional torque caused by both internal and external disturbances.

The state variables $x_i \in \text{SO}(3)$, where x_1 and x_2 are defined as

$$\begin{cases} x_1 = R, \\ x_2 = R \cdot \text{sk}(\omega_b). \end{cases} \quad (11)$$

Thus, the system can be expressed as

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = f(x) + g(x) \text{sk}(\tau_c) + D(x, d), \end{cases} \quad (12)$$

where

$$\begin{cases} f(x) = x_2 x_1^T x_2 - x_1 J_b^{-1} \text{sk} \left(x_1^T x_2 J_b (x_1^T x_2)^\vee \right), \\ g(x) = x_1 J_b^{-1}, \\ D(x, d) = x_1 J_b^{-1} \text{sk}(d). \end{cases} \quad (13)$$

The extended state $x_3 = D(x, d)$ can therefore be chosen, and then, based on the bandwidth method [42], the expression for the third-order ESO is designed as

$$\begin{cases} \dot{\hat{x}}_1 = l_1 (x_1 - \hat{x}_1) + \hat{x}_2, \\ \dot{\hat{x}}_2 = l_2 (x_1 - \hat{x}_1) + \hat{x}_3 + f(x) + g(x) \text{sk}(\tau_c), \\ \dot{\hat{x}}_3 = l_3 (x_1 - \hat{x}_1), \end{cases} \quad (14)$$

where l_i are the gain matrices, which can be expressed as

$$l_i = \alpha_i \omega_0^i I, \quad (15)$$

where I is the 3×3 identity matrix, ω_0 represents the controller bandwidth with $\omega_0 > 0$, and α_i denotes the coefficients of the $(n+1)$ -th order characteristic polynomial. In this case, since the system is of third order, the expression for α_i is given by

$$\alpha_i = \frac{3!}{i!(3-i)!}. \quad (16)$$

Let the observation error be $\tilde{x}_i = x_i - \hat{x}_i$, then

$$\begin{cases} \dot{\tilde{x}}_1 = \tilde{x}_2 - \alpha_1 \omega_0 \tilde{x}_1, \\ \dot{\tilde{x}}_2 = \tilde{x}_3 - \alpha_2 \omega_0^2 \tilde{x}_1, \\ \dot{\tilde{x}}_3 = \dot{D}(x, d) - \alpha_3 \omega_0^3 \tilde{x}_1. \end{cases} \quad (17)$$

For convenience in subsequent linearization, introduce the error variable matrix after scaling transformation

$$\begin{cases} \varepsilon = \frac{1}{\omega_0^{i-1}} \tilde{x}_i, \\ \varepsilon = [\varepsilon_1, \varepsilon_2, \varepsilon_3]^T. \end{cases} \quad (18)$$

Therefore, (17) can be expressed as

$$\dot{\varepsilon} = \omega_0 A \varepsilon + B \frac{\dot{D}(x, d)}{\omega_0^2}, \quad (19)$$

where A and B are represented as

$$A = \begin{bmatrix} -\alpha_1 I & I & 0 \\ -\alpha_2 I & 0 & I \\ -\alpha_3 I & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ I \end{bmatrix}. \quad (20)$$

Theorem 1: Assuming $\dot{D}(x, d)$ is bounded, there exist a constant $\sigma_i > 0$ and a finite $T_1 > 0$ such that $\|\varepsilon_i(t)\|_F \leq \sigma_i, \forall t \geq T_1$. Furthermore, $\sigma_i = o(\frac{1}{\omega_0^k})$, for some positive integer k .

Proof: Solving (19), it follows that

$$\varepsilon(t) = e^{\omega_0 A t} \varepsilon(0) + \int_0^t e^{\omega_0 A(t-\tau)} B \frac{\dot{D}(x(\tau), d)}{\omega_0^2} d\tau. \quad (21)$$

Let the last term in (21) be abbreviated as $p(t)$, since $\dot{D}(x, d)$ is bounded, that is $\|\dot{D}(x(\tau), d)\|_F \leq \delta$, where δ is a positive constant. Therefore

$$\begin{aligned} \|p_i(t)\|_F &\leq \frac{\delta}{\omega_0^2} \left\| \int_0^t \left(e^{\omega_0 A(t-\tau)} B \right)_i d\tau \right\|_F \\ &= \frac{\delta}{\omega_0^3} \left\| (A^{-1} e^{\omega_0 A t} B)_i - (A^{-1} B)_i \right\|_F \\ &\leq \frac{\delta}{\omega_0^3} \left(\left\| (A^{-1} e^{\omega_0 A t} B)_i \right\|_F + \left\| (A^{-1} B)_i \right\|_F \right), \end{aligned} \quad (22)$$

where A^{-1} can be expressed as

$$A^{-1} = \begin{bmatrix} 0 & 0 & -\frac{1}{\alpha_3}I \\ I & 0 & -\frac{\alpha_1}{\alpha_3}I \\ 0 & I & -\frac{\alpha_2}{\alpha_3}I \end{bmatrix}. \quad (23)$$

Let

$$v = \max \left\{ \frac{1}{\alpha_3}, \frac{\alpha_1}{\alpha_3}, \frac{\alpha_2}{\alpha_3} \right\}. \quad (24)$$

Then

$$\|(A^{-1}B)_i\|_F \leq \sqrt{3}v. \quad (25)$$

Since A is Hurwitz matrix, e^{At} will eventually tend to zero as $t \rightarrow \infty$. Since ω_0 is a positive number, $\omega_0 A$ is also a Hurwitz matrix, and a larger ω_0 results in faster convergence. After a finite time T_1 , the Frobenius norm $\|(e^{At})_{i,j}\|_F$ of each block matrix in e^{At} can be effectively bounded within an arbitrarily small limit. That is, there exists a time T_1 such that for all $t \geq T_1$

$$\|(e^{\omega_0 At})_{i,j}\|_F \leq \frac{1}{\omega_0^3}. \quad (26)$$

From the form of matrix B in (20), it can be obtained

$$\|(e^{\omega_0 At}B)_i\|_F \leq \frac{1}{\omega_0^3}. \quad (27)$$

Let

$$\mu = \max \left\{ \frac{1}{\alpha_3}, 1 + \frac{\alpha_1}{\alpha_3}, 1 + \frac{\alpha_2}{\alpha_3} \right\}. \quad (28)$$

Thus

$$\|(A^{-1}e^{\omega_0 At}B)_i\|_F \leq \frac{\mu}{\omega_0^3}. \quad (29)$$

Substituting (25) and (29) into (22)

$$\|p_i(t)\|_F \leq \frac{\sqrt{3}\delta v}{\omega_0^3} + \frac{\delta\mu}{\omega_0^6}. \quad (30)$$

Then analyze the first term in (21). From (26), it can be deduced

$$\|(e^{\omega_0 At}\mathcal{E}(0))_i\|_F \leq \frac{1}{\omega_0^3} \sum_{i=1}^3 \|\mathcal{E}_i(0)\|_F. \quad (31)$$

Substituting (30) and (31) into (21)

$$\|\mathcal{E}_i(t)\|_F \leq \frac{1}{\omega_0^3} \sum_{i=1}^3 \|\mathcal{E}_i(0)\|_F + \frac{\sqrt{3}\delta v}{\omega_0^3} + \frac{\delta\mu}{\omega_0^6}. \quad (32)$$

Restoring the scaling transformation, it can be obtained

$$\|\tilde{x}_i(t)\|_F \leq \frac{1}{\omega_0^3} \sum_{i=1}^3 \|\tilde{x}_i(0)\|_F + \frac{\sqrt{3}\delta v}{\omega_0^{4-i}} + \frac{\delta\mu}{\omega_0^{7-i}} = \sigma_i. \quad (33)$$

□

Thus, the observation error of the ESO is bounded, and its upper bound monotonically decreases with ω_0 . Although this conclusion is derived under the assumption that $\dot{D}(x, d)$ is bounded, which has certain theoretical limitations, the ESO remains effective in suppressing such disturbances since aircraft in real flight environments typically do not encounter severe impulse-form disturbances.

3.2. Design of BKS controller

Let the desired attitude matrix be R_d , and the corresponding desired body angular velocity trajectory be ω_d . The rotational error relative to the observed state in SO(3) can be expressed as

$$R_e = R_d^T R. \quad (34)$$

Its derivative is

$$\dot{R}_e = R_e \text{sk}(\omega_b) - \text{sk}(\omega_d) R_e. \quad (35)$$

A Lyapunov function can be constructed as

$$v_1 = \text{tr}(1 - R_e). \quad (36)$$

The time derivative of v_1 is

$$\dot{v}_1 = -\text{tr}(\dot{R}_e). \quad (37)$$

Substituting (35) into (37),

$$\dot{v}_1 = \text{tr}(\text{sk}(\omega_d) R_e - R_e \text{sk}(\omega_b)). \quad (38)$$

Let the coefficient matrix k_1 be a positive definite diagonal matrix. When $\text{sk}(\omega_b)$ approaches $R_e^T \text{sk}(\omega_d) R_e - k_1(R_e - R_e^T)$, it satisfies

$$\begin{aligned} \dot{v}_1 &= \text{tr}(R_e k_1 (R_e - R_e^T)) \\ &= \text{tr}(k_1 (R_e R_e - I)). \end{aligned} \quad (39)$$

To facilitate the subsequent derivation, the rotational error is further quantified by defining the rotational error vector as

$$\chi_e = (R_e - R_e^T)^\vee. \quad (40)$$

Using the properties of the trace of a matrix, it can be derived

$$\begin{aligned} \chi_e^T \chi_e &= -\frac{1}{2} \text{tr}(\text{sk}(\chi_e) \text{sk}(\chi_e)) \\ &= -\frac{1}{2} \text{tr}((R_e - R_e^T)(R_e - R_e^T)) \\ &= -\frac{1}{2} \text{tr}(R_e R_e + R_e^T R_e^T - 2I) = \text{tr}(I - R_e R_e). \end{aligned} \quad (41)$$

Therefore, from (39), it can be obtained

$$\dot{v}_1 \leq -k_{1m} \chi_e^T \chi_e \leq 0, \quad (42)$$

where k_{1m} represents the smallest element in the matrix k_1 .

Then, the tracking error of $\text{sk}(\omega_b)$ is

$$\text{sk}(z) = \text{sk}(\omega_b) - R_e^T \text{sk}(\omega_d) R_e + k_1(R_e - R_e^T). \quad (43)$$

By applying the $(\cdot)^\vee$ operator to both sides of (43), it can be derived

$$z = \omega_b - R_e^T \omega_d + k_1 \chi_e. \quad (44)$$

The derivative of the tracking error is

$$\dot{z} = \dot{\omega}_b - (\dot{R}_e^T \omega_d + R_e^T \dot{\omega}_d) + k_1 \dot{\chi}_e, \quad (45)$$

where

$$\dot{\omega}_b = J_b^{-1}(\tau_c - \omega_b \times J_b \omega_b). \quad (46)$$

Therefore, the Lyapunov function is further designed as

$$v_2 = v_1 + \frac{1}{2} z^T z. \quad (47)$$

The time derivative of v_2 is

$$\dot{v}_2 = \dot{v}_1 + z^T \dot{z}. \quad (48)$$

For the first term in (48), substituting (43) into (38), it can be obtained

$$\dot{v}_1 \leq -k_{1m} \chi_e^T \chi_e - \text{tr}(R_e \text{sk}(z)). \quad (49)$$

Using the properties of the trace of a matrix, it can be derived

$$\begin{aligned} z^T \chi_e &= -\frac{1}{2} \text{tr}(\text{sk}(z) \text{sk}(\chi_e)) \\ &= -\frac{1}{2} \text{tr}(\text{sk}(z) R_e - \text{sk}(z) R_e^T) \\ &= -\frac{1}{2} \text{tr}(\text{sk}(z) R_e + \text{sk}(z)^T R_e^T) \\ &= -\text{tr}(\text{sk}(z) R_e). \end{aligned} \quad (50)$$

Substituting (50) into (49),

$$\dot{v}_1 \leq -k_{1m} \chi_e^T \chi_e + z^T \chi_e. \quad (51)$$

Substituting (51) and (45) into (48),

$$\begin{aligned} \dot{v}_2 &\leq -k_{1m} \chi_e^T \chi_e + z^T \chi_e + z^T k_1 \dot{\chi}_e \\ &\quad + z^T (J_b^{-1}(\tau_c - \omega_b \times J_b \omega_b) - (\dot{R}_e^T \omega_d + R_e^T \dot{\omega}_d)). \end{aligned} \quad (52)$$

Thus, the controller is designed as

$$\begin{aligned} \tau_c &= \omega_b \times J_b \omega_b \\ &\quad + J_b(-\chi_e + \dot{R}_e^T \omega_d + R_e^T \dot{\omega}_d - k_1 \dot{\chi}_e - k_2 z), \end{aligned} \quad (53)$$

where the coefficient matrix k_2 is a positive definite diagonal matrix, and k_{2m} represents the smallest element in the matrix k_2 . Hence, it can be obtained:

$$\dot{v}_2 \leq -k_{1m} \chi_e^T \chi_e - k_{2m} z^T z \leq 0. \quad (54)$$

When and only when $\chi_e = 0$ and $z = 0$, $\dot{v}_2 = 0$ is satisfied. According to the LaSalle invariance principle, as $t \rightarrow \infty$, $\chi_e \rightarrow 0$ and $z \rightarrow 0$, which results in $R_e \rightarrow I$, and thus $R \rightarrow R_d$.

3.3. Stability analysis of closed-loop system

To analyze the stability of the entire closed-loop system, the Lyapunov function is designed as follows:

$$v_3 = \text{tr}(I - R_e) + \frac{1}{2} z^T z + \frac{1}{2} \text{tr}(\varepsilon^T P \varepsilon), \quad (55)$$

where P is the unique positive definite matrix that satisfies

$$A^T P + P A = -I. \quad (56)$$

Let the last term of (55) be q . Based on the properties of the matrix trace, it can be obtained

$$q \geq \frac{1}{2} \lambda_{\max}(P) \|\varepsilon\|_F^2. \quad (57)$$

Its derivative is

$$\begin{aligned} \dot{q} &= \frac{1}{2} \text{tr}(\dot{\varepsilon}^T P \varepsilon + \varepsilon^T P \dot{\varepsilon}) \\ &= \frac{1}{2} \text{tr} \left(\omega_0 \varepsilon^T A^T P \varepsilon + \omega_0 \varepsilon^T P A \varepsilon + 2 \varepsilon^T P B \frac{\dot{D}(x, d)}{\omega_0^2} \right) \\ &= -\frac{1}{2} \text{tr}(\omega_0 \varepsilon^T \varepsilon) + \text{tr} \left(\varepsilon^T P B \frac{\dot{D}(x, d)}{\omega_0^2} \right) \\ &= -\frac{\omega_0}{2} \|\varepsilon\|_F^2 + \text{tr} \left(\varepsilon^T P B \frac{\dot{D}(x, d)}{\omega_0^2} \right). \end{aligned} \quad (58)$$

According to the Cauchy-Schwarz inequality and the RMS-arithmetic mean inequality

$$\begin{aligned} \text{tr} \left(\varepsilon^T P B \frac{\dot{D}(x, d)}{\omega_0^2} \right) &\leq \|\varepsilon\|_F \left\| P B \frac{\dot{D}(x, d)}{\omega_0^2} \right\|_F \\ &\leq \frac{1}{2} \|\varepsilon\|_F^2 + \frac{1}{2} \left\| P B \frac{\dot{D}(x, d)}{\omega_0^2} \right\|_F^2 \\ &\leq \frac{1}{2} \|\varepsilon\|_F^2 + \frac{\delta^2}{2 \omega_0^4} \|P\|_F^2. \end{aligned} \quad (59)$$

Substituting (59) into (58)

$$\dot{q} \leq \frac{1 - \omega_0}{2} \|\varepsilon\|_F^2 + \frac{\delta^2}{2 \omega_0^4} \|P\|_F^2. \quad (60)$$

Let

$$\zeta = \frac{\delta^2}{2 \omega_0^4} \|P\|_F^2. \quad (61)$$

The derivative of the Lyapunov function satisfies

$$\dot{v}_3 \leq -k_{1m} \chi_e^T \chi_e - k_{2m} z^T z - \frac{\omega_0^{-1}}{2} \|\varepsilon\|_F^2 + \zeta. \quad (62)$$

According to (57), let

$$\lambda = \min\{2k_{1m}, 2k_{2m}, \frac{\omega_0 - 1}{\lambda_{\max}(P)}\}. \quad (63)$$

Then

$$\dot{v}_3 \leq -\lambda v_3 + \zeta. \quad (64)$$

This differential equation can be solved as

$$v_3(t) \leq e^{-\lambda t} v_3(0) + \frac{\zeta}{\lambda} (1 - e^{-\lambda t}). \quad (65)$$

Therefore, the designed Lyapunov function is bounded. Furthermore, all signals in the closed-loop system are bounded, and the entire control system is boundedly stable.

4. VALIDATION AND DISCUSSION

4.1. Simulation results

To verify the effectiveness and robustness of the designed controller, a series of simulation experiments were conducted in MATLAB. The physical parameters of the drone used in the simulation are as follows: $m = 0.37$ kg, $J_b = \text{diag}\{2916, 1593, 1856\}$ kgmm², $L_i = \text{diag}\{110, 95, 110\}$ mm, $m_i^p = 0.037$ kg, $r_i^p = 22$ mm. To better approximate real-world physical conditions in the simulation, angular velocity feedback signals were superimposed with random noise of $\pm 0.05^\circ/\text{s}$ amplitude and subjected to a time delay equivalent to one control cycle. The controller frequency in the simulation environment is set to 100 Hz, with the initial attitude defined as $\phi = 0, \theta = 0, \psi = 0$. To eliminate the influence of manual parameter tuning, a PSO algorithm [13] was employed to optimize the controller parameters. The PSO parameter settings are specified in Table 1, with the corresponding flowchart illustrated in Fig. 3.

Four different control methods were designed for the comparative experiment, as follows:

Table 1. Parameter settings of the PSO algorithm.

PSO Parameters and result	Value
Number of particles	50
Number of iteration	50
Inertia weight	0.7
Cognitive coefficient	1.5
Social coefficient	1.5

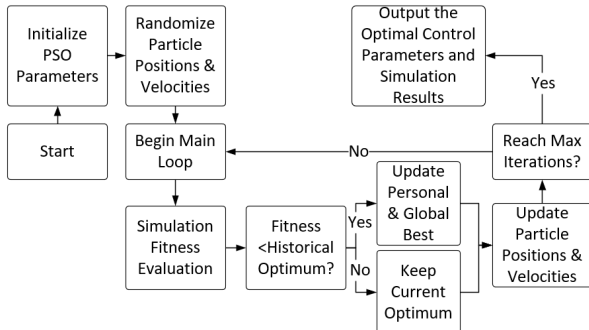


Fig. 3. Flowchart of the PSO algorithm.

- PID controller based on separated Euler angles (denoted as PID in Euler). This is the most commonly used control method for drones.
- PID in SO(3). The design method is based on the research by Yu *et al.* [25] and He *et al.* [13].
- BKS in SO(3).
- BKS & ESO in SO(3).

The performances of the four controllers under different operating conditions are compared below, with the expected signals being numerically coupled across multiple rotational axes. The related simulation data are shown in Figs. 4-17. The animation used for demonstration can be found at <https://youtu.be/G0AsXOzsEIQ>.

1) Small angle step signal

The step command for the desired signal occurs at 0.5 s, with the values set as $\phi = 0.5$, $\theta = 0.5$, and $\psi = 0.5$. The response characteristics of the different controllers to the desired signal are shown in Fig. 4, the control signals are shown in Fig. 5. The Euler angle-based control method fails to guarantee smooth response characteristics across all rotational axes due to inter-axis coupled moments and insufficient fine parameter tuning for distinct axes. Increasing target control parameters via the PSO algorithm might alleviate this issue but would substantially

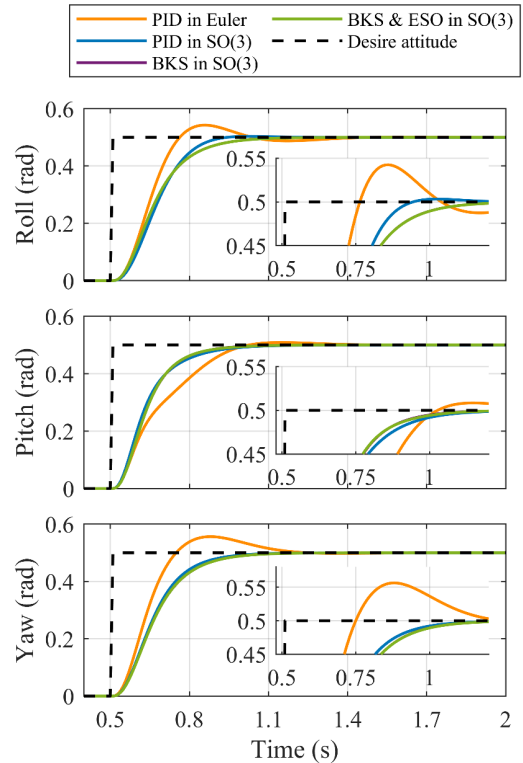


Fig. 4. Response characteristics under small angle step signal.

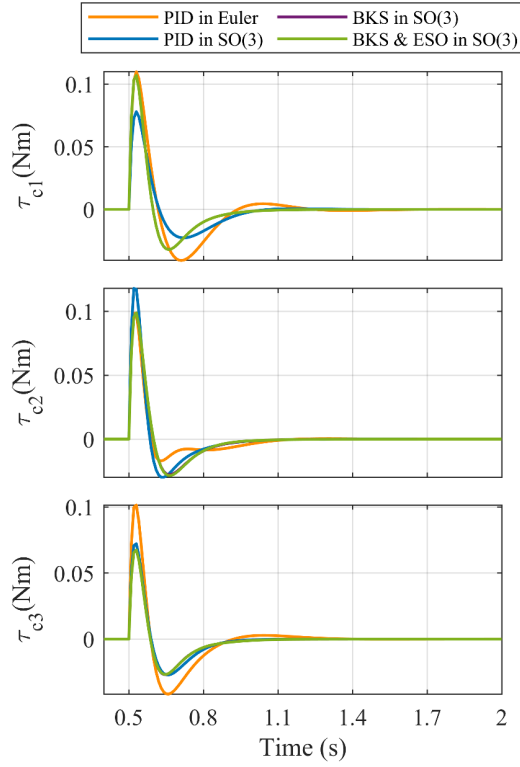


Fig. 5. Control signals under small angle step responses.

elevate computational demands. In contrast, all three controllers designed on SO(3) rapidly approach desired values within 0.5 s, exhibiting smooth curves without overshoot and demonstrating excellent response characteristics. As no disturbances were introduced, the combined BKS and ESO scheme exhibits identical response characteristics to the standalone BKS controller, with their curves overlapping completely.

2) Large angle step signal

The values for the desired signal are changed to: $\phi = 2.0$, $\theta = 2.0$, and $\psi = 2.0$. Due to the singularities and domain limitations inherent in Euler angle representations, these factors are not conducive to displaying data. Therefore, results presented in Figs. 6 and 7, and subsequent data are presented in the form of errors in the Lie algebra, with the related calculation methods following the research of Yu *et al.* [25]. In Fig. 6, it can be observed that when the value of any rotation axis exceeds $\frac{\pi}{2}$, the Euler angle-based control method loses control rapidly due to issues like gimbal lock and the influence of inter-axis coupling moments, resulting in severe error oscillations and sustained divergence. In contrast, the three controllers designed in SO(3) do not face these problems and can quickly stabilize.

3) Large angle step signal with eccentric load

To test the disturbance rejection capabilities of each

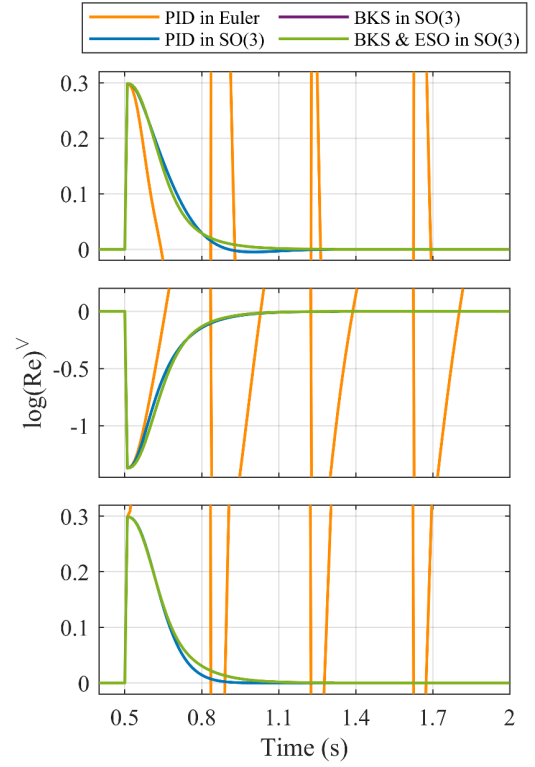


Fig. 6. Tracking error in Lie algebra under large angle step signal.

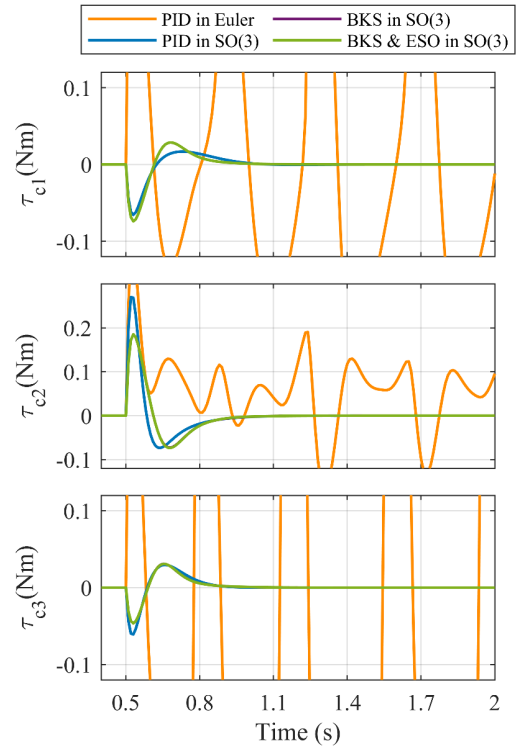


Fig. 7. Control signals under large angle step responses.

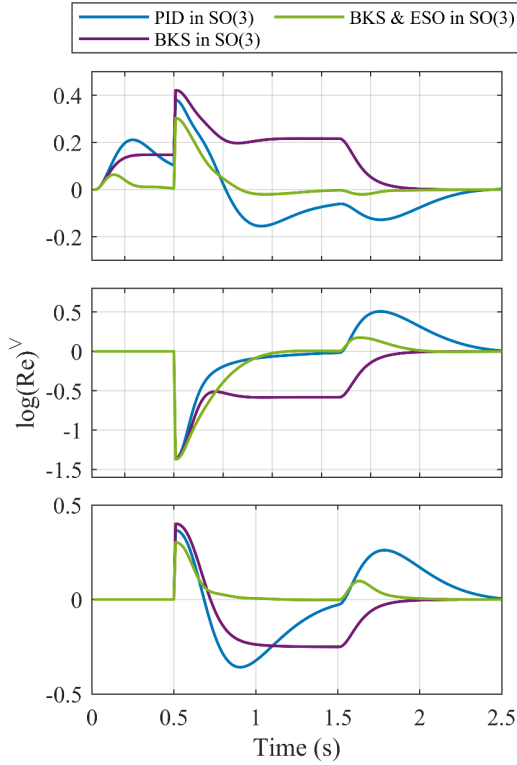


Fig. 8. Tracking error in Lie algebra under large angle step signal with eccentric load.

controller, an eccentric disturbance model was established in the simulation environment based on the bound equivalent mass particle. Its expression is as follows:

$$d = m_d r_d \times (\dot{\omega}_b \times r_d + g), \quad (66)$$

where m_d is the mass of the equivalent particle (0.1 kg), and r_d is the vector representing the eccentric position ($[0.0, 0.1, -0.2]$ m). To substantiate the reasonableness of the bounded disturbance derivative assumption adopted in deriving (33), the eccentric load mass was suddenly set to zero at 1.5 s in the simulation, emulating the scenario where an aircraft releases an eccentric payload during material delivery missions.

The performance of the three controllers designed in $SO(3)$ under this condition is compared, and the results are shown in Figs. 8 and 9. The standalone BKS controller cannot eliminate steady-state errors. The PID controller, with the help of the integral term, can gradually drive the tiltrotor's attitude toward the desired values, but this process is markedly slower than the combined BKS and ESO scheme. All control parameters employed were optimized via the PSO algorithm, thereby eliminating any influence from subjective parameter tuning. The demonstrated performance of the PID controller represents the optimal solution achievable without inducing significant overshoot. In contrast, the combination of BKS and ESO allows the

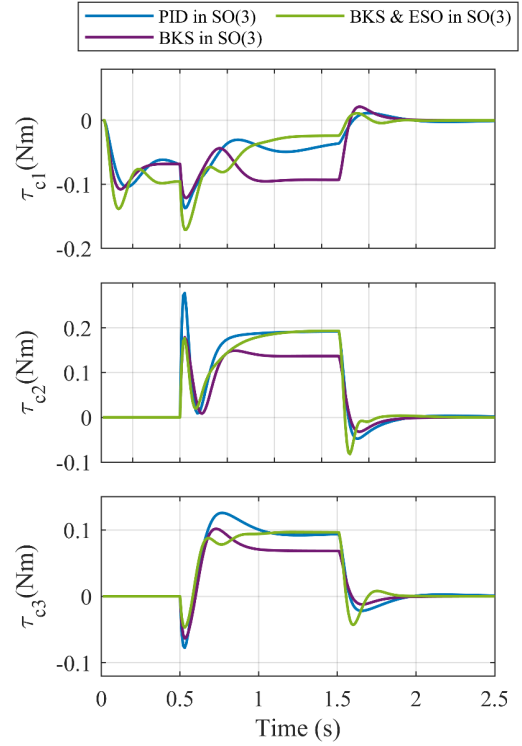


Fig. 9. Control signals under large angle step signal with eccentric load.

steady-state error to quickly converge to zero, exhibiting excellent robustness. Even under conditions involving sudden disturbance changes, the proposed control method maintains rapid convergence. Fig. 10 demonstrates the observer's performance in estimating unknown disturbances during this process, revealing its ability to rapidly and accurately predict disturbance conditions.

4) Low-speed rotational trajectory tracking

The desired signals are defined as $\phi = 0.5 \sin(0.5t)$, $\theta = 2.0 \sin(0.5t)$, and $\psi = 0.5 \cos(0.5t)$. As shown in Figs. 11 and 12, the tracking error of the PID controller for time-varying signals is significantly larger than that of the other two controllers, especially in the pitch axis. This is because the amplitude of the desired pitch angle is the largest, amplifying this phenomenon.

5) High-speed rotational trajectory tracking

The desired signals are given as $\phi = 2.0 \sin(2.0t)$, $\theta = 2.0 \sin(2.0t)$, and $\psi = 2.0 \cos(2.0t)$. The data trends in Figs. 13 and 14 are very similar to those in Figs. 11 and 12, but with larger numerical values. The provided demonstration animation offers a clearer visualization, showing that the tracking error of the PID controller is significant. In contrast, the other two controllers exhibit only minor errors, even under high-speed rotational conditions.

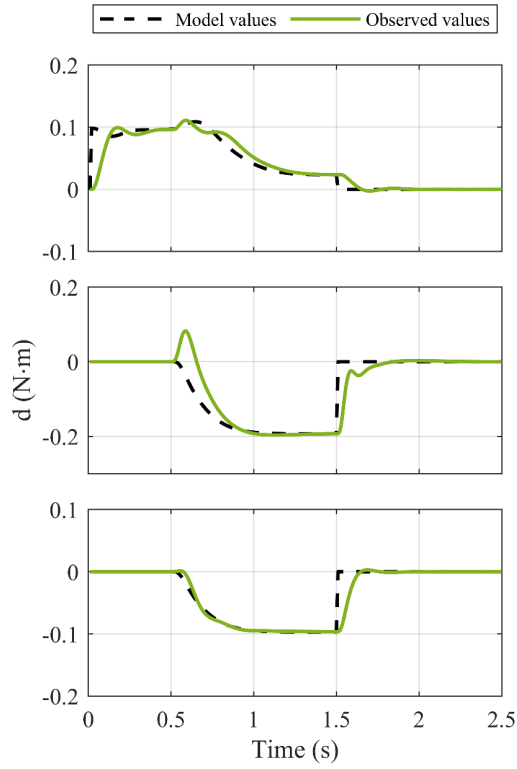


Fig. 10. Disturbance observation performance under large angle step signal with eccentric load.

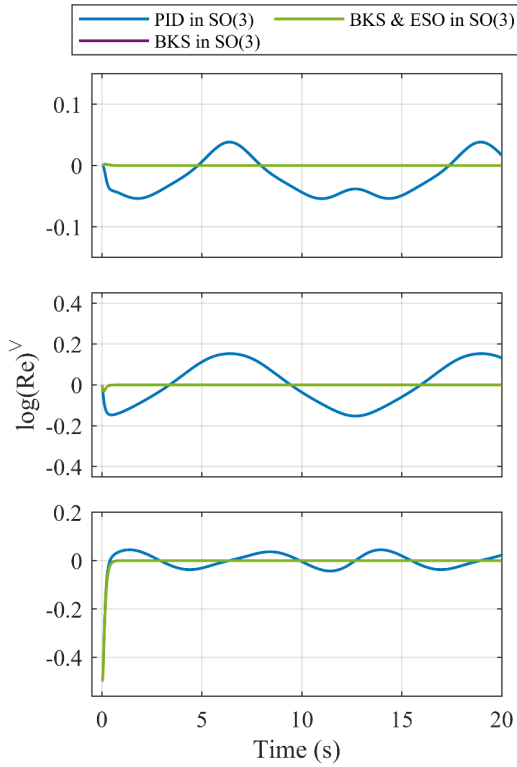


Fig. 11. Tracking error in Lie algebra for low-speed rotation trajectory.

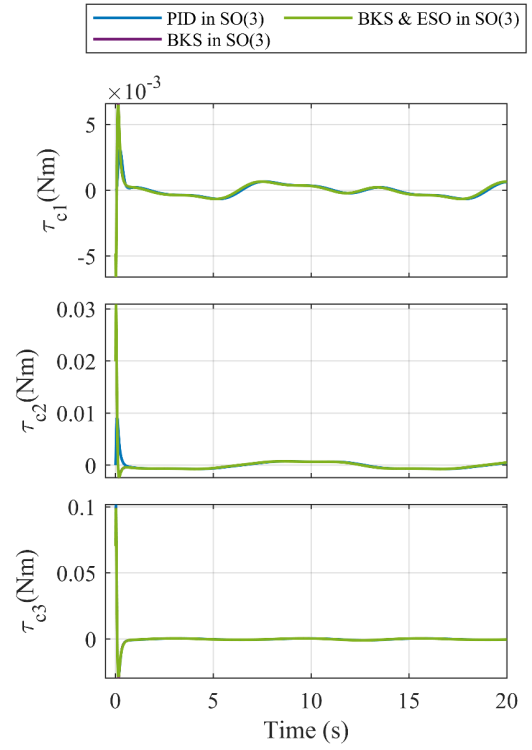


Fig. 12. Control signals for low-speed rotation trajectory.

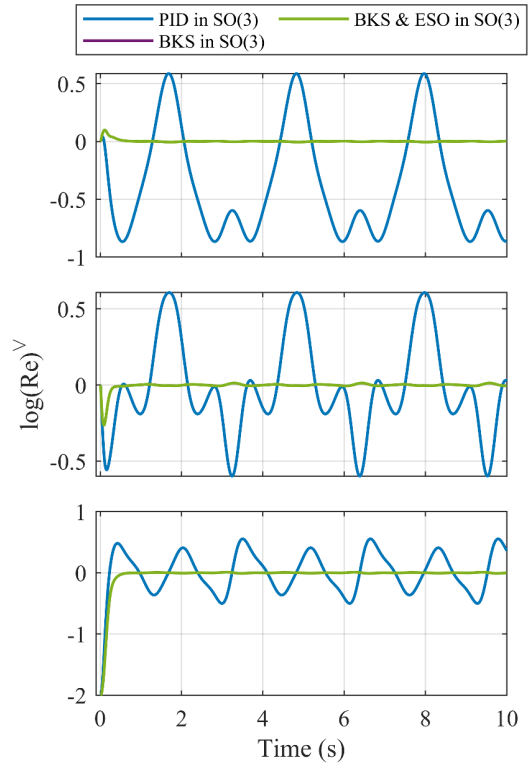


Fig. 13. Tracking error in Lie algebra for high-speed rotation trajectory.

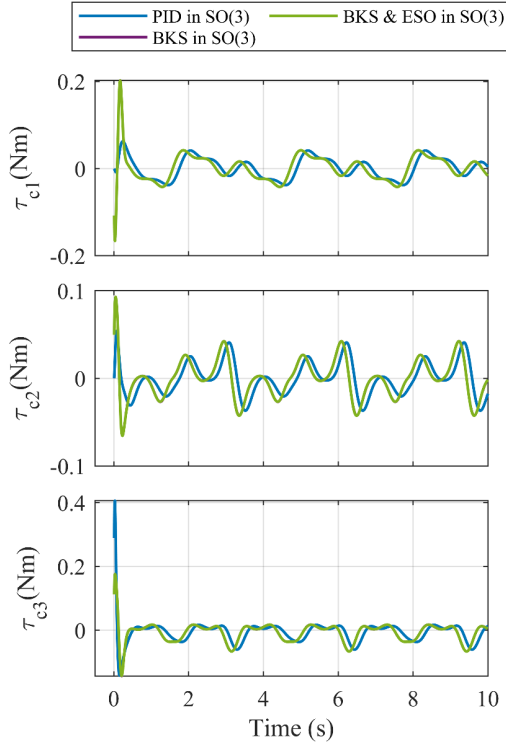


Fig. 14. Control signals for high-speed rotation trajectory.

6) High-speed rotational trajectory tracking with eccentric load

The disturbance model remains the same as in (66). From Figs. 15 and 16, it is evident that the PID controller exhibits the largest tracking error, while the standalone BKS controller also shows noticeable errors. Only the combination of BKS and ESO maintains a consistently low tracking error. Fig. 17 illustrates the disturbance observation results of the ESO, showing that the observed values closely match the model data. This verifies the accuracy of the ESO and confirms that the controller designed in SO(3) in this study is both effective and robust.

4.2. Prototype testing platform and discussion

The bi-copter prototype, designed based on the 3D model shown in Fig. 1, has been manufactured. Additionally, to validate the attitude control method proposed in this study, a simple two-axis testing platform has been constructed, as shown in Fig. 18.

The onboard controller is designed using the STM32F405RGT6 microcontroller. The control algorithm was implemented using DSP libraries, with a single-loop execution time of approximately $260 \mu\text{s}$, demonstrating low computational resource requirements sufficient to support high-frequency control demands. The main control frequency and IMU sampling frequency are both set at 100 Hz, with a data downlink rate of 20 Hz. The bi-copter

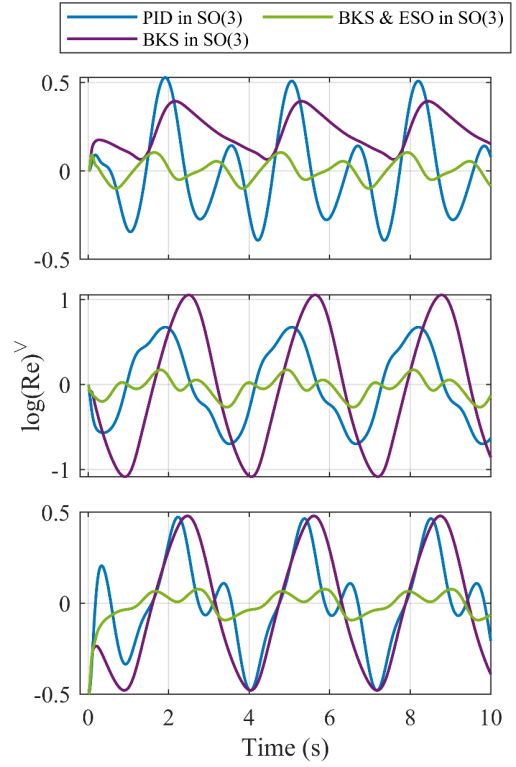


Fig. 15. Tracking error in Lie algebra for high-speed rotation trajectory with eccentric load.

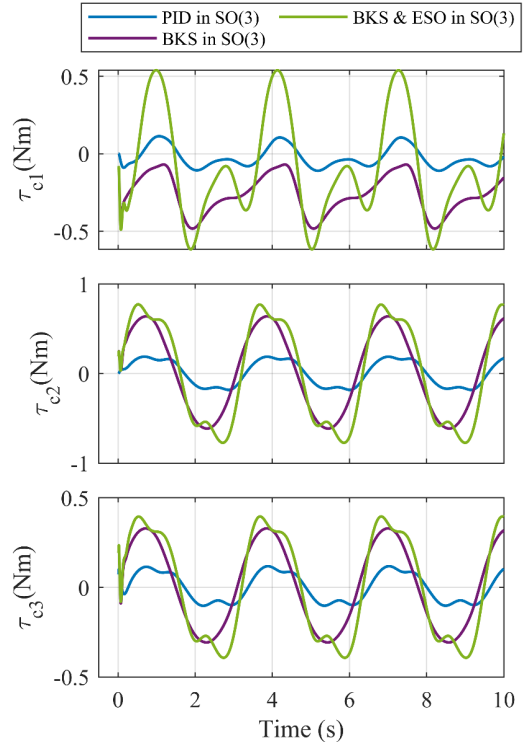


Fig. 16. Control signals for high-speed rotation trajectory with eccentric load.

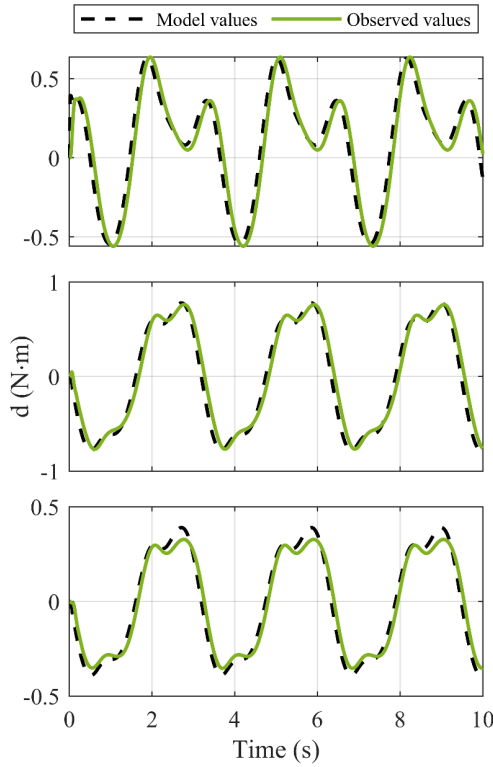


Fig. 17. Observation accuracy of disturbance for high-speed rotation trajectory with eccentric load.



Fig. 18. Bi-copter prototype and testing platform.

prototype is mounted on a carbon fiber square tube, which allows for rotational freedom along the yaw axis of the aircraft. The carbon fiber square tube is then mounted on an aluminum alloy bracket, providing rotational freedom along its central axis. This setup allows for the verification of the controlled behavior of the bi-copter prototype under two-axis rotational conditions.

Fig. 19 demonstrates the tracking performance of the bi-copter prototype in response to the multi-axis high-speed rotational reference signals, and Fig. 20 displays the corresponding control signals, representing the de-

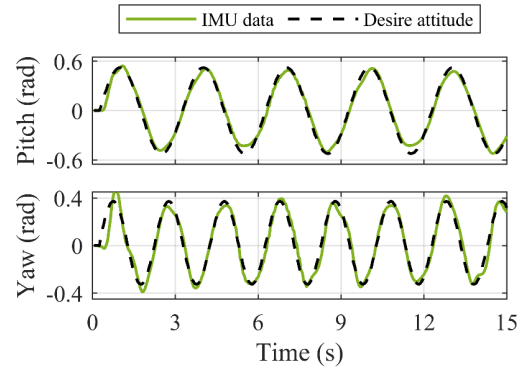


Fig. 19. Bi-copter prototype and testing platform.

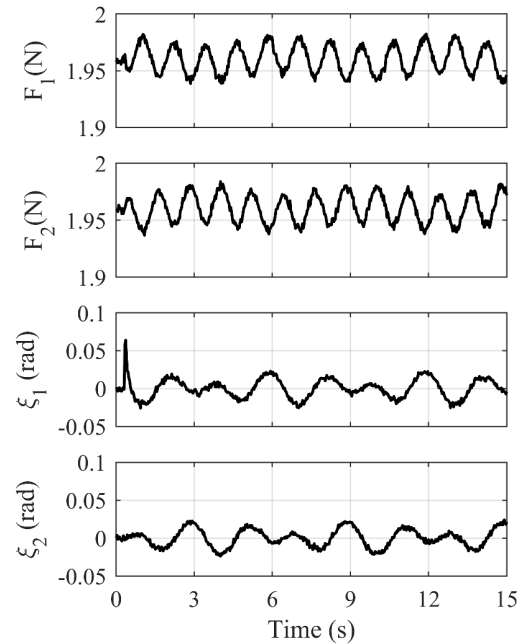


Fig. 20. Bi-copter prototype and testing platform.

sired propeller thrust and micro servo actuator deflection angle, respectively. The reference signals are given as $\phi = \frac{\pi}{6} \sin(\frac{2\pi}{3}t)$, $\psi = \frac{\pi}{9} \sin(\pi t)$. It can be seen that the designed controller accurately tracks the multi-axis coupled reference signals for most of the time. However, there are some instances of significant tracking errors, such as the pitch angle data around 5.5 seconds and 11.5 seconds. Our analysis reveals that this is not due to a flaw in the controller design, but rather a limitation of the simple testing platform. When the Euler angles of the bi-copter prototype are significantly away from the hover point, the rotational axis of the carbon fiber square tube of the testing platform does not align with any of the aircraft's pitch, roll, or yaw axes. Since all the control quantities in the bi-copter dynamics (1) are aligned with the aircraft's own rotational axes, the moment generated by the eccentric

weight cannot be balanced by any controller on the bi-copter prototype. This results in temporary controller malfunctions, especially when the multi-axis Euler angles of the prototype are large. Nonetheless, even under these adverse conditions, the designed controller is able to maintain bounded stability, validating its strong robustness.

In addition, due to the underactuated characteristics of the bi-copter, flight experiments involving complex maneuvers cannot guarantee that the gravitational forces will always be balanced, which is why actual flight experiments have not been conducted yet. In the future, we plan to develop a new type of drone with vertical takeoff and landing capability, based on the current bi-copter prototype. This new drone will feature special wings capable of multi-directional movement to balance the gravitational forces. The control methods proposed in this study will be further applied to this new drone, and a spatial position control component will be incorporated to achieve high-maneuverability shuttle operations in complex urban environments.

5. CONCLUSION AND FUTURE WORK

This study proposes a robust attitude controller for bi-copters in $SO(3)$, which includes the design of the corresponding BKS controller and ESO. The bounded stability of the controller is proven using the Lyapunov method. The proposed control approach allows the bi-copter's controllable attitude to extend beyond the vicinity of the hover point. This approach shows significant advantages over other control methods when dealing with disturbances and complex maneuvers. In future work, we aim to develop a new type of drone based on improvements to the bi-copter design, incorporating multiple flight modes. The control strategies proposed in this study for its complex maneuvers will be applied to this new drone, and outdoor flight experiments will be conducted to further validate the control strategies.

DECLARATIONS

Conflict of Interest

The authors declare that there is no competing financial interest or personal relationship that could have appeared to influence the work reported in this paper.

Authors' Contributions

Tianyou Mao conducted the entire manuscript writing. Bosong Duan contributed to the logical structure and format standardization of the manuscript. Chuangqiang Guo proposed the main research idea and provided funding support.

Funding

This work is supported by the National Natural Science Foundation of China (No. 52475013), the Natural Science Foundation of Heilongjiang Province (No. ZD2023E002), the Open Fund of Laboratory of Aerospace Servo Actuation and Transmission (No. LASAT-2022-A01), National Natural Science Foundation of China (No. 52405037), and Aviation Science Fund (No. 2024Z006076001).

REFERENCES

- [1] N. Papyan, M. Kulhandjian, H. Kulhandjian, and L. Aslanyan, "Ai-based drone assisted human rescue in disaster environments: Challenges and opportunities," *Pattern Recognition and Image Analysis*, vol. 34, pp. 169-186, 2024.
- [2] M. Kowshika, M. Ooviya, B. Pavithradevi, K. V. Rashika, and S. S. Kumar, "Unmanned aerial systems in search and rescue: A comprehensive review and future directions," *Proc. of 2024 5th International Conference on Mobile Computing and Sustainable Informatics (ICMCSI)*, pp. 15-18, 2024.
- [3] S. Rangan, Y. Shanmugam, G. Valluvan, and T. Prakash, "A customized drone for ocean and atmospheric measurements and its performances," *Maritime Technology and Research*, vol. 6, no. 3, 267638, 2024.
- [4] S.-Y. Wang and C.-D. Lin, "Using deep reinforcement learning to train and periodically re-train a data-collecting drone based on real-life measurements," *Journal of Network and Computer Applications*, vol. 221, 103789, 2024.
- [5] S. Ijaz, Y. Shi, Y.A. Khan, M. Khodaverdian, and U. Javaid, "Robust adaptive control law design for enhanced stability of agriculture UAV used for pesticide spraying," *Aerospace Science and Technology*, vol. 155, no. 2, 109676, 2024.
- [6] P. Kalaiselvi, J. Chaurasia, A. Krishnaveni, A. Krishnamoorthi, A. Singh, V. Kumar, S. Sapna, and R. Labanya, "Harvesting efficiency: The rise of drone technology in modern agriculture," *Journal of Scientific Research and Reports*, vol. 30, no. 6, pp. 191-207, 2024.
- [7] S. Wu, S. Shi, X. Wang, and J. Jiang, "Integral event-triggered time-varying formation tracking for UAV swarm systems with switching directed topologies," *International Journal of Control, Automation, and Systems*, vol. 22, no. 8, pp. 2624-2632, 2024.
- [8] A. Rizaldi and Y. Kim, "Optimization of dual-type multi-drone formations for load-carrying missions," *International Journal of Control, Automation, and Systems*, vol. 23, no. 2, pp. 498-509, 2025.
- [9] N. Li, H. Wang, Q. Luo, and W. Zheng, "Distributed formation control for multiple quadrotor uavs based on distributed estimator and singular perturbation system," *International Journal of Control, Automation, and Systems*, vol. 22, no. 4, pp. 1349-1359, 2024.
- [10] Y. Qin, W. Xu, A. Lee, and F. Zhang, "Gemini: A compact yet efficient bi-copter UAV for indoor applications," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 3213-3220, 2020.

- [11] H. Gong, B. Huang, B. Jia, and H. Dai, "Modeling power consumptions for multirotor UAVs," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 6, pp. 7409-7422, 2023.
- [12] M. Taherinezhad, A. Ramirez-Serrano, and A. Abedini, "Robust trajectory-tracking for a bi-copter drone using INDI: A gain tuning multi-objective approach," *Robotics*, vol. 11, no. 5, pp. 86, 2022.
- [13] X. He, and Y. Wang, "Design and trajectory tracking control of a new bi-copter UAV," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 9191-9198, 2022.
- [14] Q. Zhang, Z. Liu, J. Zhao, and S. Zhang, "Modeling and attitude control of Bi-copter," *Proc. of IEEE International Conference on Aircraft Utility Systems (AUS)*, pp. 172-176, 2016.
- [15] Y. Li, Y. Qin, W. Xu, and F. Zhang, "Modeling, identification, and control of non-minimum phase dynamics of bi-copter UAVs," *Proc. of IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, pp. 1249-1255, 2020.
- [16] S. Li, F. Liu, Y. Gao, J. Xiang, Z. Tu, and D. Li, "AirTwins: Modular bi-copters capable of splitting from their combined quadcopter in midair," *IEEE Robotics and Automation Letters*, vol. 8, no. 9, pp. 6068-6075, 2023.
- [17] K. Shi, Z. Jiang, L. Ma, L. Qi, and M. Jin, "MTABot: An efficient morphable terrestrial-aerial robot with two transformable wheels," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1875-1882, 2024.
- [18] S. Verling, B. Weibel, M. Boosfeld, K. Alexis, M. Burri, and R. Siegwart, "Full attitude control of a VTOL tailsitter UAV," *Proc. of IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3006-3012, 2016.
- [19] J. M. O. Barth, J.-P. Condomines, J.-M. Moschetta, A. Cabarbaye, C. Join, and M. Fliess, "Full model-free control architecture for hybrid UAVs," *Proc. of American Control Conference (ACC)*, pp. 71-78, 2019.
- [20] N. A. Chaturvedi, A. K. Sanyal, and N. H. McClamroch, "Rigid-body attitude control," *IEEE Control Systems Magazine*, vol. 31, no. 3, pp. 30-51, 2011.
- [21] E. A. Basso, H. M. Schmidt-Didlauskies, and K. Y. Petersen, "Global asymptotic position and heading tracking for multirotors using tuning function-based adaptive hybrid feedback," *IEEE Control Systems Letters*, vol. 7, pp. 295-300, 2023.
- [22] J. Yang, Z. Cai, J. Zhao, Z. Wang, Y. Ding, and Y. Wang, "INDI-based aggressive quadrotor flight control with position and attitude constraints," *Robotics and Autonomous Systems*, vol. 159, pp. 104292, 2023.
- [23] C. Montañez-Molina and J. Pliego-Jiménez, "Position and reduced attitude trajectory tracking control of quadrotors: Theory and experiments," *Aerospace Science and Technology*, vol. 155, no. 2, pp. 109683, 2024.
- [24] J. Madeiras, C. Cardeira, and P. Oliveira, "Trajectory tracking of a quadrotor via full-state feedback control," *Aerospace Science and Technology*, vol. 147, 109013, 2024.
- [25] Y. Yu, S. Yang, M. Wang, C. Li, and Z. Li, "High performance full attitude control of a quadrotor on SO(3)," *Proc. of IEEE International Conference on Robotics and Automation (ICRA)*, pp. 1698-1703, 2015.
- [26] E. Apriaskar, F. Fahmizal, N. A. Salim, and D. Prastiyanto, "Performance evaluation of balancing bicopter using P, PI, and PID controller," *Jurnal Teknik Elektro*, vol. 11, no. 2, pp. 44-49, 2019.
- [27] A. Jayadi, F.D. Adhinata, J. P. Sembiring, N. U. Putri, E. Pranita, and A. Setiawan, "Implementation of PID control as a one axis balancing control system on the bicopter vehicle," *Proc. of 2023 International Conference on Converging Technology in Electrical and Information Engineering (ICCTEIE)*, pp. 113-117, 2023.
- [28] Ö. B. Albayrak, Y. Ersan, A. S. Bağbaşı, A. Turgut Başaranoglu, and K.B. Arıkan, "Design of a robotic bicopter," *Proc. of 7th International Conference on Control, Mechatronics and Automation (ICCMA)*, pp. 98-103, 2019.
- [29] E. Apriaskar, D. Prastiyanto, A. A. Manaf, I. Amelia, and Fahmizal, "Multi-criteria genetic algorithm optimization approach for balancing bicopter control," *IOP Conference Series: Earth and Environmental Science*, vol. 1203, 012027, 2023.
- [30] E. Apriaskar, D. Prastiyanto, A. B. Utomo, A. A. Manaf, I. Amelia, D. A. Ilham, V. L. Bilqis, C. Photong, E. Apriaskar, D. Prastiyanto, A. B. Utomo, A.A. Manaf, I. Amelia, D.A. Ilham, V.L. Bilqis, and C. Photong, "Genetic algorithm-enhanced linear quadratic control for balancing bicopter system with non-zero set point," *Journal of Mechatronics, Electrical Power, and Vehicular Technology*, vol. 14, pp. 105-113, 2023.
- [31] J. A. Prakosa, E. Kurniawan, Purwowibowo, and C. Astuti, "Design of robust H-infinity flight control for MIMO model of bi-copter against uncertainty," *Proc. of International Symposium on Electronics and Smart Devices (IS-ESD)*, pp. 1-6, 2022.
- [32] A. N. Jadoon, A. Mughees, I. Ahmad, and H. I. Sherazi, "Optimal novel nonlinear control law for attitude and heading control of a bi-copter system with controller-in-loop," *Expert Systems With Applications*, vol. 272, pp. 126554, 2025.
- [33] A. Abedini, A. A. Bataleblu, and J. Roshanian, "Robust backstepping control of position and attitude for a bi-copter drone," *Proc. of 2021 9th RSI International Conference on Robotics and Mechatronics (ICRoM)*, pp. 425-432, 2021.
- [34] Z.-Q. Hu, C.-C. Hua, Y.-N. Yang, and L.-L. Zhang, "Adaptive finite-time control of quadrotor uavs with asymmetric input saturation," *International Journal of Control, Automation, and Systems*, vol. 23, no. 1, pp. 286-299, 2025.
- [35] M. Cao, H. Wang, F. Ma, B. Zhu, P. Ji, and H. Zhang, "Non-fragile finite-time contractive control of a quadrotor for trajectory tracking," *International Journal of Control, Automation, and Systems*, vol. 23, no. 1, pp. 274-285, 2025.
- [36] P. J. D. de Oliveira Evald, V. M. Aoki, C. B. da Silva, D. S. Cardoso, P. M. Pinheiro, S. S. da C. Botelho, and P. L. J. Drews Jr., "A review on quadrotor attitude control strategies," *International Journal of Intelligent Robotics and Applications*, vol. 8, pp. 230-250, 2024.

- [37] H. Bai, S. Mei, and J. Dong, "Hierarchical temporal sequence convergence in prescribed-time control for quadrotor UAVs under unknown dynamic disturbances," *Aerospace Science and Technology*, vol. 148, pp. 109094, 2024.
- [38] Z. Li, S. Liang, M. Guo, H. Zhang, H. Wang, Z. Li, and H. Li, "ADRC-based underwater navigation control and parameter tuning of an amphibious multirotor vehicle," *IEEE Journal of Oceanic Engineering*, vol. 49, no. 3, pp. 1-18, 2024.
- [39] S. Shen, J. Xu, and Q. Xia, "A fuzzy backstepping attitude control based on an extended state observer for a tilt-rotor UAV," *Aerospace*, vol. 9, no. 11, pp. 724, 2022.
- [40] N. Dalwadi, D. Deb, and S. Ozana, "Dual observer based adaptive controller for hybrid drones," *Drones*, vol. 7, no. 1, pp. 48, 2023.
- [41] Z. Gao, "Scaling and bandwidth-parameterization based controller tuning," *Proc. of American Control Conference*, pp. 4989-4996, 2003.
- [42] Q. Zheng, L.Q. Gaol, and Z. Gao, "On stability analysis of active disturbance rejection control for nonlinear time-varying plants with unknown dynamics," *Proc. of 2007 46th IEEE Conference on Decision and Control*, pp. 3501-3506, 2007.



Tianyou Mao is currently working toward a Ph.D. degree at the State Key Laboratory of Robotics and Systems, Harbin Institute of Technology. His research interests include flapping-wing aircraft, multi-rotor aircraft, and vertical take-off and landing aircraft.



Bosong Duan received his Ph.D. degree from Harbin Institute of Technology, China, in 2023. He is a research assistant in Zhejiang University. His current research interests include FAV design and atomic-level manufacturing.



Chuangqiang Guo is a Research Fellow and Doctoral Supervisor at the State Key Laboratory of Robotics and Systems, Harbin Institute of Technology. His research focuses on space robotic system design and control, on-orbit fault detection and fault tolerance for space manipulators, and micro flapping-wing aircraft design and control.